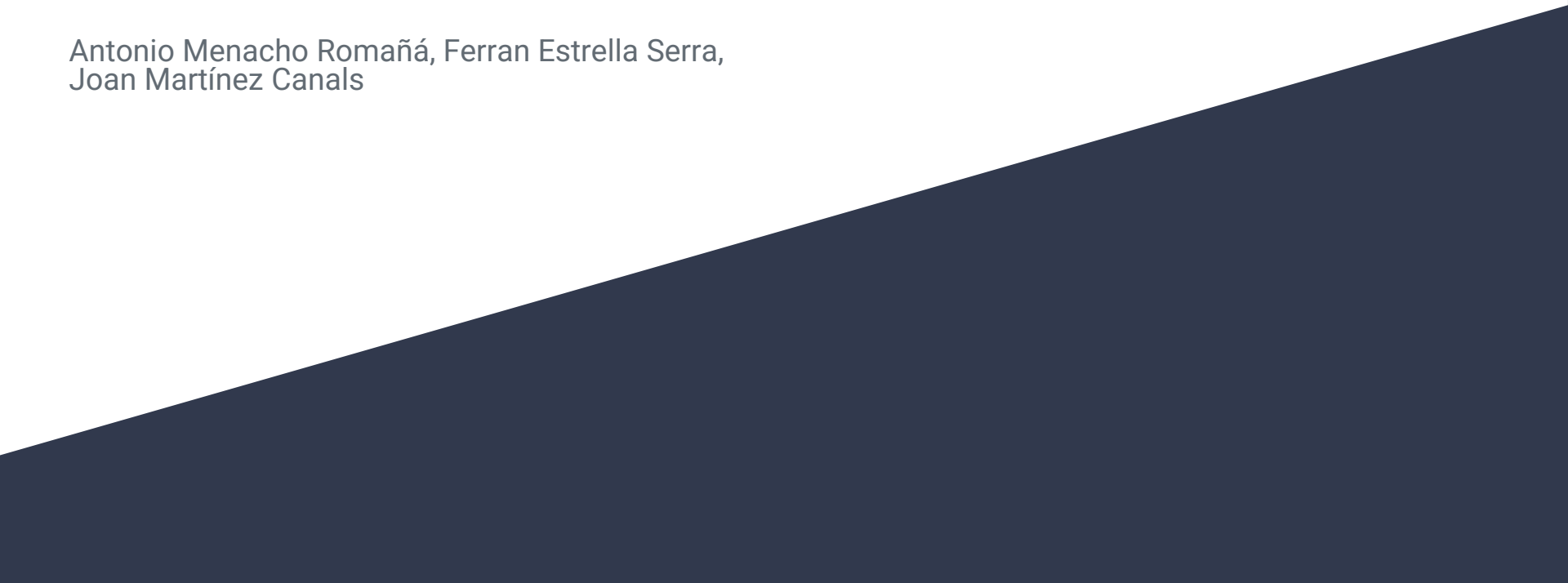


Bayesian methods to predict global life expectancy

Antonio Menacho Romañá, Ferran Estrella Serra,
Joan Martínez Canals

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Motivation

Predictive Complexity

Life expectancy and other global sociological markers are notoriously difficult to predict.

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Structural Progress

It serves as the ultimate proxy for assessing structural progress across nations.

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The Macroeconomic Mirror

A complex proxy reflecting the intersection of healthcare, economics, and inequality.

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The Dataset Overview

Source

Kaggle's WHO Life Expectancy

Observations

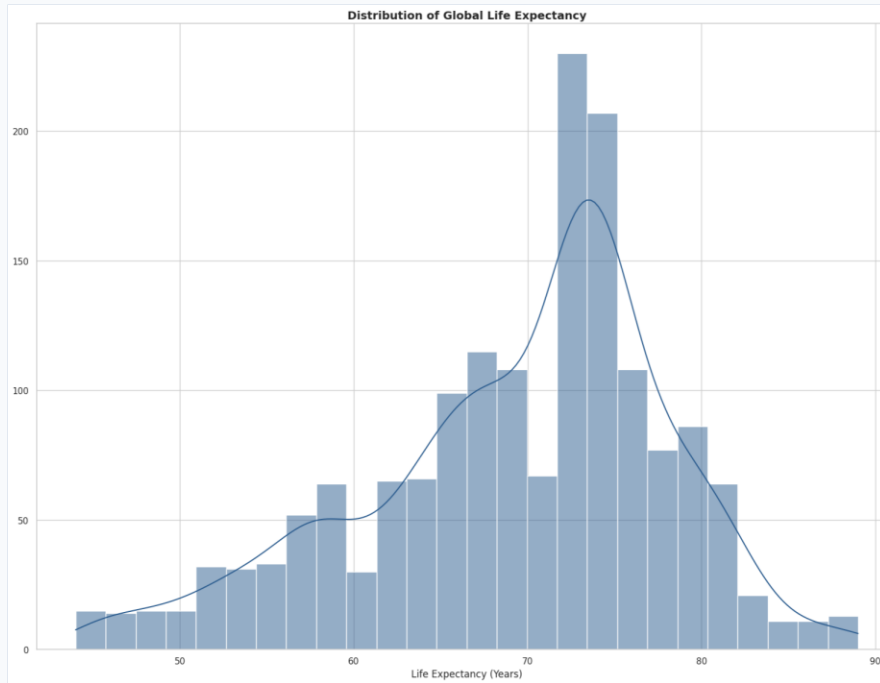
2,938

Countries

193

21 Input Variables

- **Immunization:** Polio, Hepatitis B...
- **Disease:** Measles, HIV/AIDS
- **Economic:** GDP, Health Expenditure
- **Social:** Schooling, Human Dev.



Source: <https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who>

Hostile data environment

- We chose this particular dataset because it is **notoriously bad**
- Exploratory analysis reveals: errors in reporting, data leakage, and redundant variables.
- Ill-conditioned data is common in real word socio-economic contexts. Hence we wanted a complex, noisy environment to “stress test” our models.
- Moreover, exploratory analysis of the data reveals **two inherent structural challenges** that provide meaningful comparison for our models

Challenge I: Multicollinearity

This dataset involves redundant features that create mathematical tension in standard models. Variables like infant death and under-five deaths have a correlation of 1

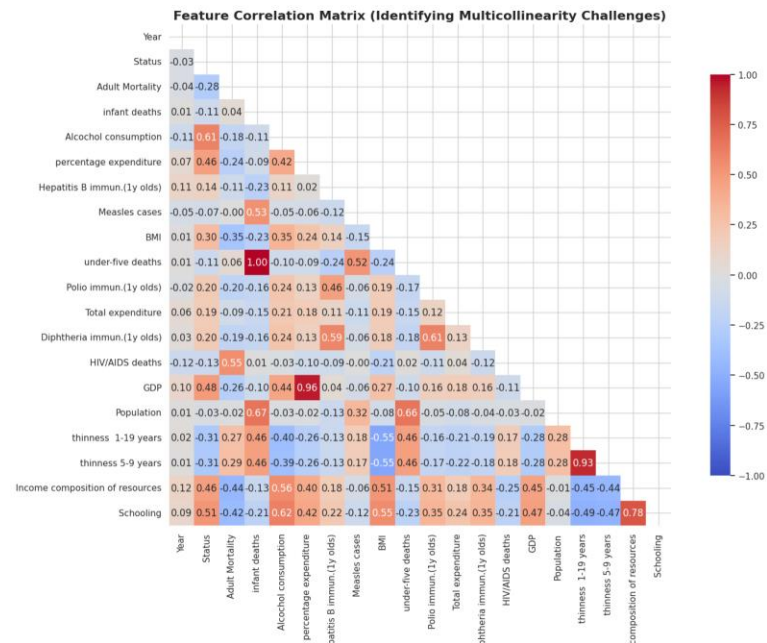
High Redundancy

Infant & Under-Five Deaths: **1.0 correlation**

OLS Limitation

OLS will likely struggle as it cannot hold one variable constant while changing the other.

Creates a meaningful environment to compare with **Bayesian Ridge Regression** stabilizes weights via **Gaussian priors**.



Challenge II: Non-linear Dynamics

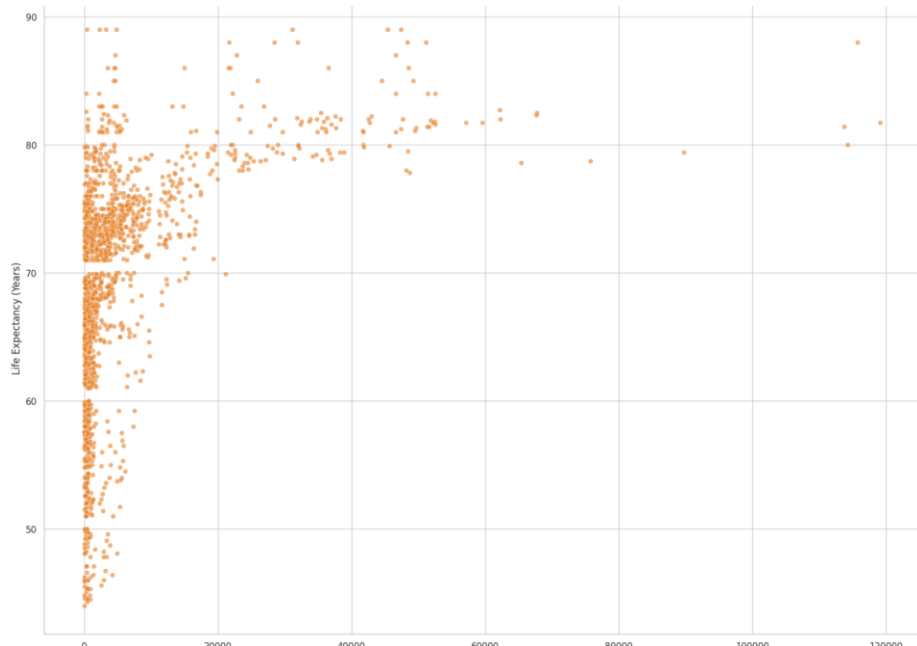
Diminishing Returns of Wealth

Economic factors like GDP per capita exhibit severe curves: wealth causes a survival spike initially but hits a biological plateau at higher tiers.

Non-Linear Dynamics

This dataset allows for a meaningful comparison between:

- **Linear Bayesian Ridge Regression**
- **Kernel Ridge Regression (KRR):**
Maps predictors into higher-dimensional spaces.



Data preprocessing

Removing Invalid data

- % entries with values >100
- $x/1000$ entries with values >1000
- Removing entries with null values

Targeting macro trends

We drop the countries column

- We blind the model to the country the data comes from
- Force it to look at macroeconomic trends

Categorical Abstraction

Encode categorical values

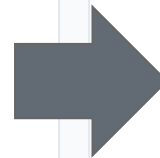
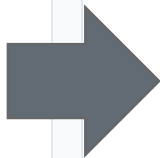
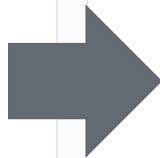
We encode the status variable:

- Developed: 1
- Developing: 0

Feature standardization

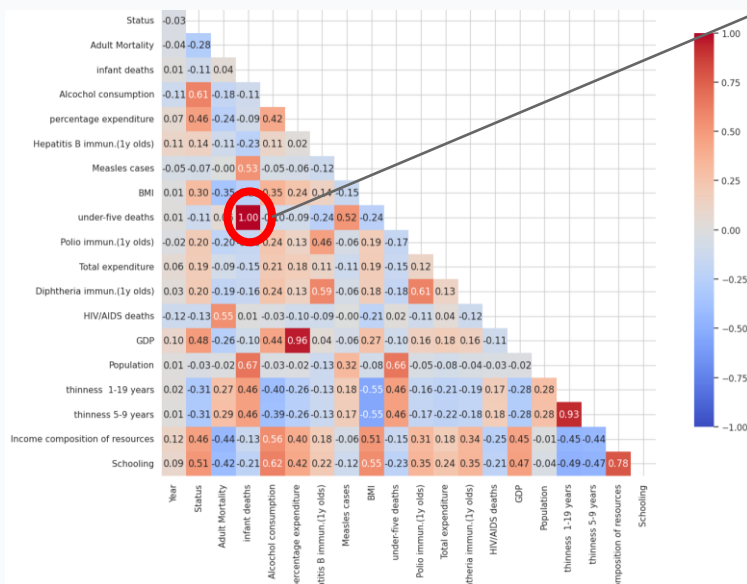
Variable values exist in very different scales.

- We use z-score scaling so that:
- Mean is 0
- Std dev is 1



Methodology: Bayesian Ridge Experiments

Experiment 1: OLS and Bayesian Ridge vs collinearity



VERY Collinear or even redundant (similar, if not the same, definition!)

Will lead to **PROBLEMS** in OLS, Expect to be **'FIXED'** by Bayesian Ridge.

Methodology: Plot **OLS** (used a Bayesian Ridge with $\lambda \ll 1$) and **Bayesian Ridge** ($\lambda \gg 1$) trained **WITH** and **WITHOUT**:
'under-five deaths' data.

Methodology: Bayesian Ridge Experiments

Experiment 2: Bayesian Ridge Prior Precision's impact on weights

- KEEP comparison between with and without 'under-five deaths' data.
- Use Ridge to be able to control the prior precision.
- Use $\lambda \in [10^{-1}, 10^5]$. To plot weight shrinkage.

Bayesian Ridge (sklearn):

Updates λ on each iteration → NO STRICT CONTROL ON PRIOR PRECISION

$$\lambda = \frac{\sigma^2}{\tau^2} \rightarrow 1$$

Ridge (sklearn):

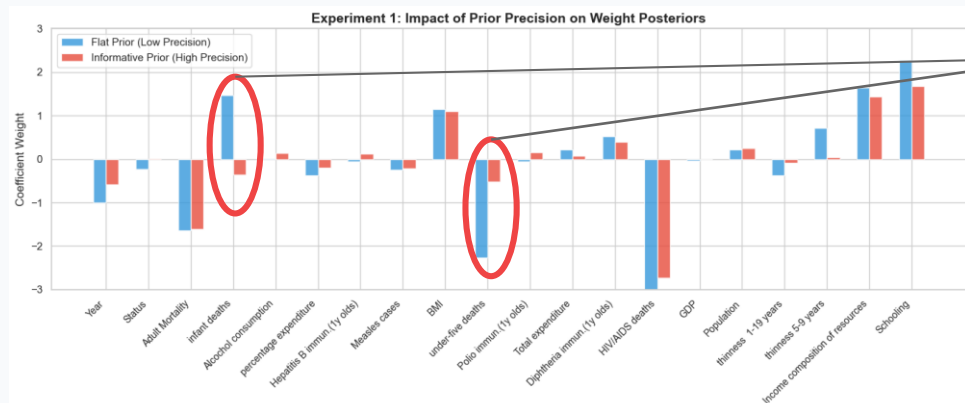
Uses given λ for training → STRICT CONTROL ON PRIOR PRECISION

Since:

$$L(\mathbf{w}) = \text{RSS}(\mathbf{w}) + \lambda \|\mathbf{w}\|_2^2$$

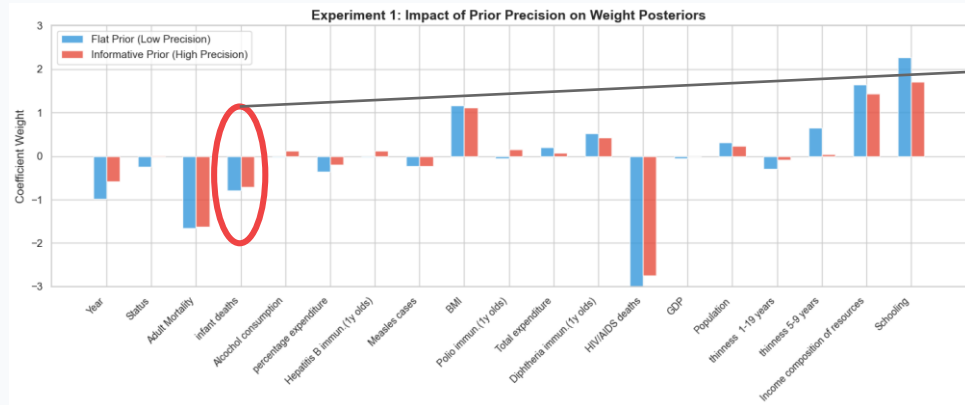
We expect a SHRINKAGE in the norm of the weights by INCREASING λ .

Results: Bayesian Ridge vs Collinearity



OLS: More infant deaths increase life expectancy, as long as under-five deaths decrease.

WITH under-five deaths

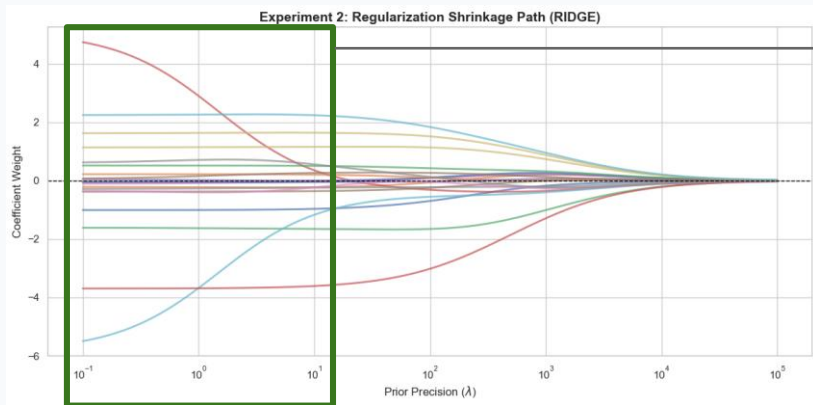


OLS: Avoids that illogical stance and has similar weights to Bayesian Ridge!

Bayesian Ridge is able to compensate this collinearity!

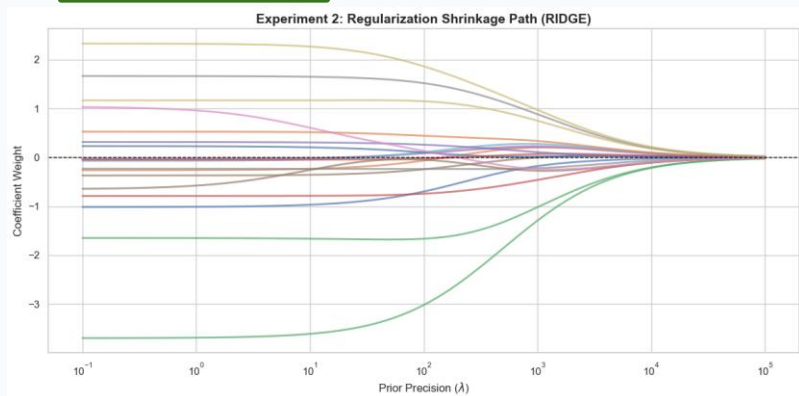
WITHOUT under-five deaths

Results: Bayesian Ridge vs Collinearity



Quick compensation of troubling pair!

WITH under-five deaths



WITHOUT under-five deaths

Methodology: Kernel Ridge Regression (KRR)

Optimization problem:

$$\min_{f \in \mathcal{H}} \frac{1}{2} \sum_{i=1}^N (y_i - f(x_i))^2 + \frac{\alpha}{2} \|f\|_{\mathcal{H}}^2$$

$\alpha \equiv$ penalization parameter

- Representer theorem:

$$f(x) = \sum_{j=1}^N c_j K(x_j, x)$$

- RBF Kernel:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

$\gamma \equiv$ length-scale parameter

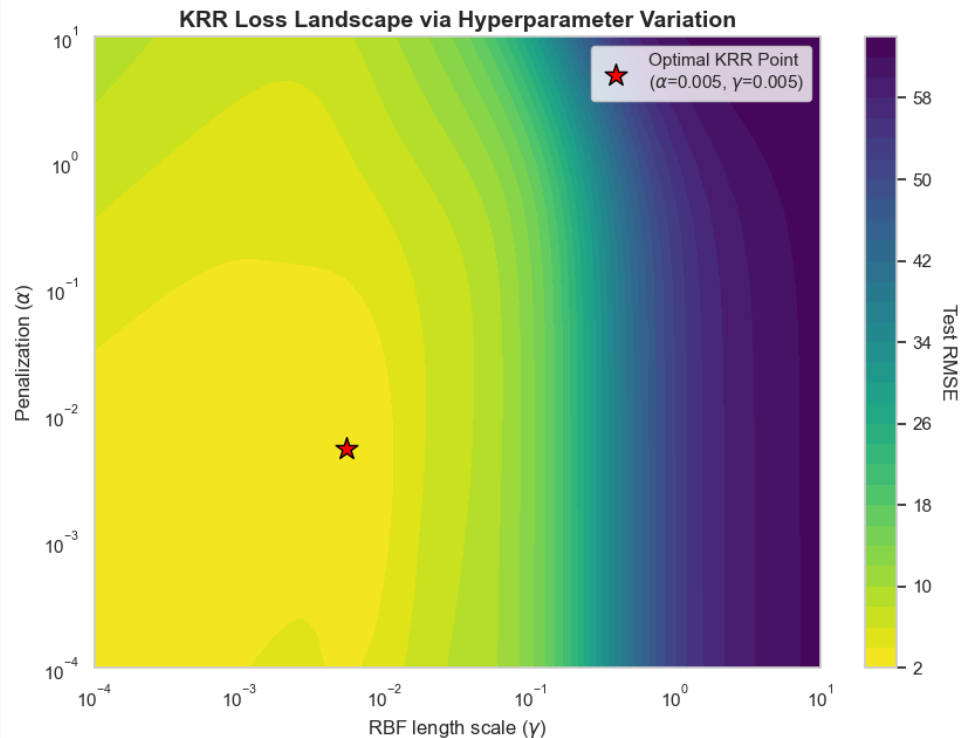
Equivalent problem:

$$\min_{\mathbf{c} \in \mathbb{R}^N} \left\{ \|\mathbf{y} - K_{XX}(\gamma) \mathbf{c}\|_2^2 + \alpha \mathbf{c}^\top K_{XX}(\gamma) \mathbf{c} \right\}$$

Solution:

$$\mathbf{c} = (K_{XX}(\gamma) + \alpha I)^{-1} \mathbf{y}$$

Results: Kernel Ridge Regression (KRR)



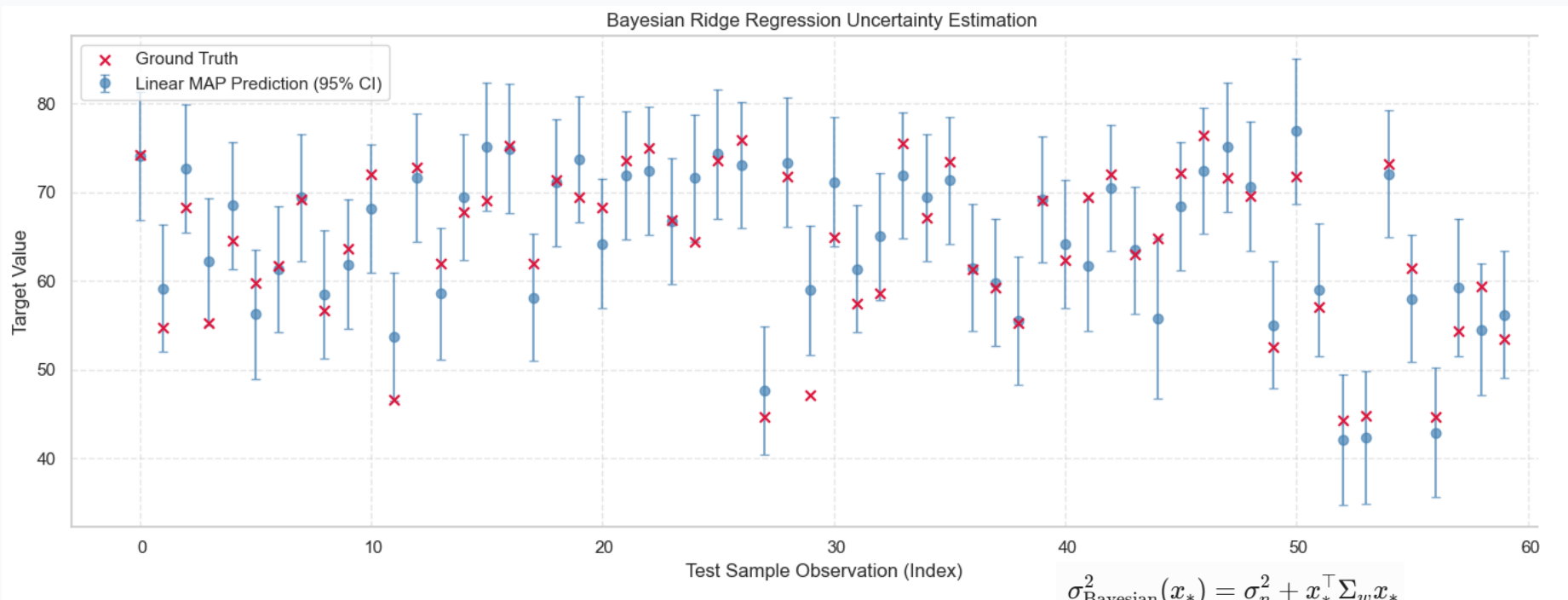
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FINAL EVALUATION METRICS (test sample)

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Optimal Bayesian Linear Ridge RMSE: 3.6708 | Avg Std Error: 3.7019
Optimal RBF KRR RMSE: 2.4371 | Avg Std Error: 1.9472

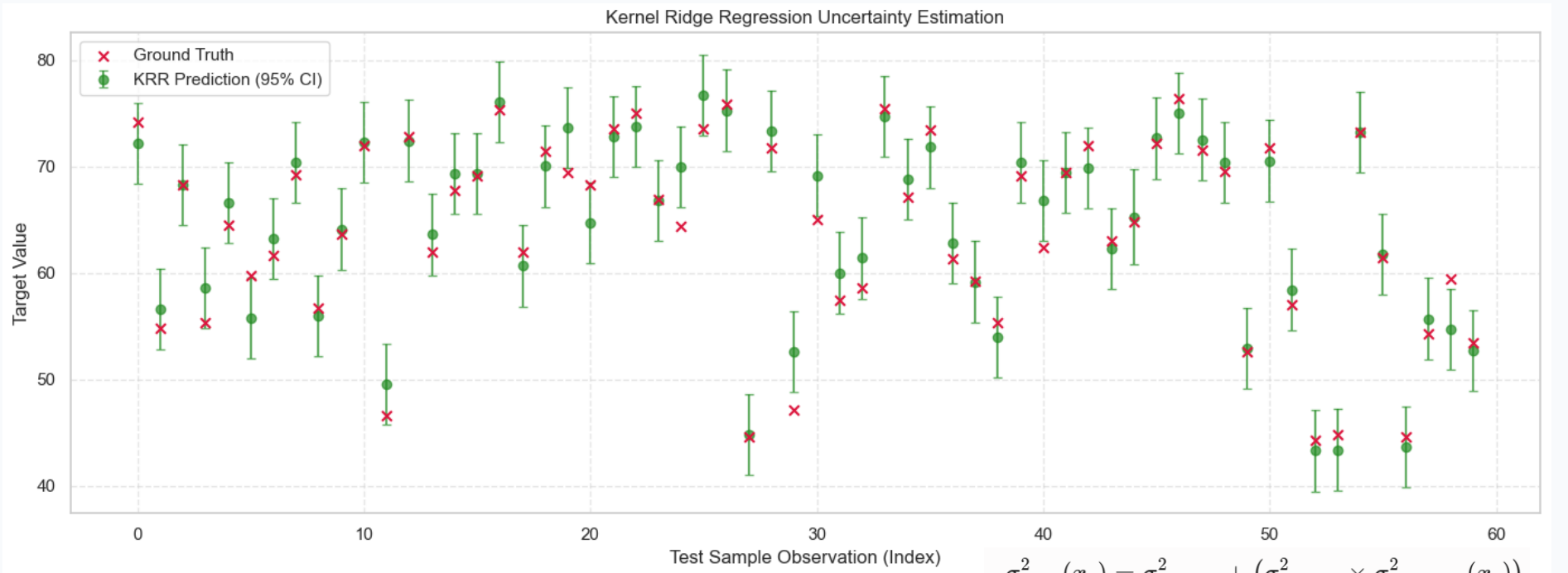
Results: Uncertainty Quantification



$$\sigma_{\text{Bayesian}}^2(x_*) = \sigma_n^2 + x_*^\top \Sigma_w x_*$$

$$\Sigma_w = \left(\frac{1}{\sigma_n^2} X_{\text{train}}^\top X_{\text{train}} + \frac{1}{\tau^2} I \right)^{-1}$$

Results: Uncertainty Quantification



$$\sigma_{\text{KRR}}^2(x_*) = \sigma_{\text{residual}}^2 + (\sigma_{\text{residual}}^2 \times \sigma_{\text{geometric}}^2(x_*))$$

$$\sigma_{\text{residual}}^2 = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

$$\sigma_{\text{geometric}}^2(x_*) = K(x_*, x_*) - \mathbf{k}_*^\top (K_{XX}(\gamma) + \alpha I)^{-1} \mathbf{k}_*$$

Conclusions

- Bayesian Ridge Regression (BRR) effectively handles correlated variables through weight shrinkage, succeeding where standard OLS fails.
- Kernel Ridge Regression (KRR) outperforms Bayesian Ridge Regression by successfully mapping the complex non-linear dynamics of socioeconomic features.
- BRR yields broad, conservative, confidence intervals due to its global linear assumptions
- KRR adapts to the local data structure and generate significantly tighter and more accurate confidence bands.

Thank you for your attention!

Feel free to ask questions!